ALGORITHM · DIVIDE AND CONQUER

MERGE SORT

A divide-and-conquer algorithm.

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Made with GAMMA

From scattered numbers to sorted order

Merge sort works efficiently on smaller pieces instead of comparing everything at once.

START · UNSORTED

38	27	43	3
9	82	10	

TARGET · SORTED

3	9	10	27
38	43	82	

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Three moves power merge sort

1

Divide

Split the array into smaller subarrays.

2

Conquer

Sort the smaller subarrays.

3

Merge

Combine them into one sorted array.

ONE ARRAY

```
[38, 27, 43, 3, 9, 82, 10]
```

ONE RESULT

```
[3, 9, 10, 27, 38, 43, 82]
```

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Divide the array, preserving order

Divide recursively to single elements; each is already sorted. Preserve left-to-right order.

```
[38, 27, 43, 3, 9, 82, 10]
```

```
[38, 27, 43, 3] + [9, 82, 10]
```

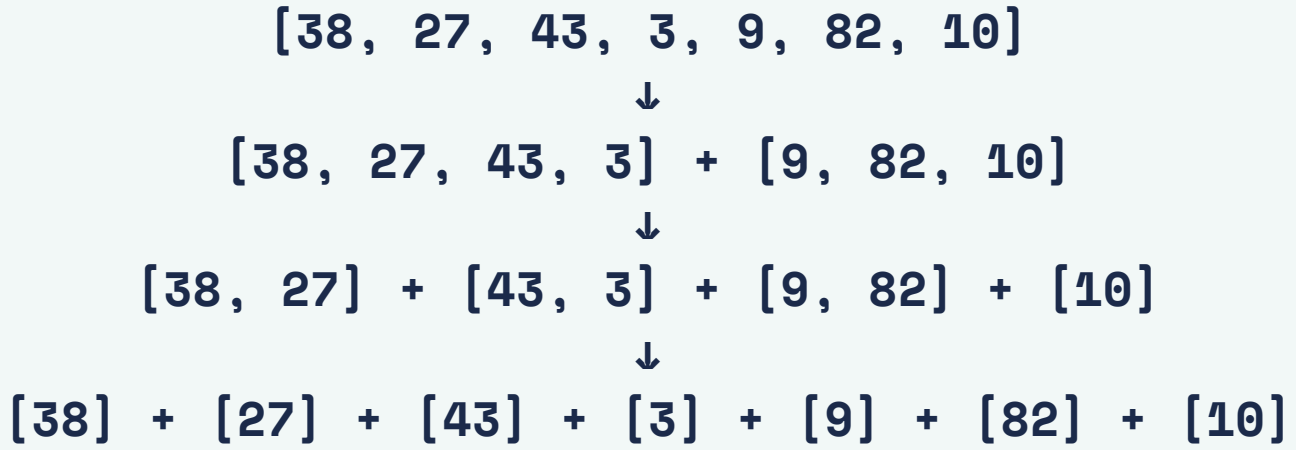
```
[38, 27] + [43, 3] + [9, 82] + [10]
```

```
[38] + [27] + [43] + [3] + [9] + [82] + [10]
```

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The split as a recursion tree

Each branch stops when it reaches one already-sorted element.



Stopping rule: one element means already sorted.

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Once divided, merging creates order

The return journey builds sorted groups from the smallest pieces.

MERGE PAIRS	MERGE GROUPS	FINAL MERGE
[27, 38] + [3, 43] + [9, 82] + [10]	[3, 27, 38, 43] + [9, 10, 82]	[3, 9, 10, 27] [38, 43, 82]

Break a hard problem into simple pieces, solve each piece, and combine the results.

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Thank you

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